Analytical Cumulative Examination
February 7, 2003

Citric acid is a triprotic acid with pK_a's of 3.13, 4.76 and 6.40, for the three successive dissociations.

a. Write the mass action expressions for the three dissociations.

b. Determine the following three ratios of the citrate species:

$A = [H_3Cit]/[H_2Cit^-]$

$B = [H_2Cit^-]/[HCit^{2-}]$

$C = [HCit^{2-}]/[Cit^{3-}]$

At pH 1, at pH 4.76, and at pH 7.00

c. Derive an expression for the fraction available (α) in the form Cit^{3-}, as a function of K_1, K_2, K_3 and $[H^+]$.

d. Calculate the fraction available in the form Cit^{3-} at pH 5.0.